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Development of digital twin system for central air-conditioning based on BIM

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ABSTRACT

With the rapid development of building intelligence, the efficient management and control of central air-conditioning systems have become a research hotspot in the HVAC (Heating, Ventilation, and Air Conditioning) field. The effective integration of BIM (Building Information Modeling) technology and digital twins into HVAC systems enables the visualization of smart operation and maintenance processes, as well as information sharing across complex equipment systems, providing technical support for modeling, simulation, fault detection, and optimized control of air conditioning systems. This paper proposes a BIM-based digital twin framework for central air-conditioning systems. Through secondary development of a laboratory air conditioning system's BIM model, a communication architecture and data-driven logic for the central air conditioning digital twin system is presented. The system is designed and developed using OPC/UA technology and the Unity platform. Comprehensive performance testing validates the stability and reliability of the digital twin platform, The test results show that, under typical operating scenarios, the platform consistently maintains a CPU frame time within 10 ms, memory usage below 2 GB, script execution time under 10 ms, and fewer than 2000 rendering batches. All functional test cases were successfully passed, demonstrating stable feature responsiveness and validating the platform's excellent performance in response speed, security, and interactive capabilities. These results provide strong technical support for the intelligent operation and maintenance of central air-conditioning systems.

1. Introduction

The emergence of new information technologies such as the Internet of Things (IoT) [1], big data [2], and artificial intelligence (AI) [3] has positioned digital twins [4] as a bridge connecting physical entities with their digital counterparts, becoming a key enabler for digital transformation and intelligent upgrades [5]. Digital twin technology utilizes IoT to collect and synchronize real-time parameter data from the physical world, enabling comprehensive information monitoring, in-depth data analysis, accurate predictive simulations, and efficient optimization control. This opens new pathways for intelligent development across numerous fields. In recent years, digital twin technology has been widely applied in energy management, building environmental control, and industrial system optimization.

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Against the backdrop of increasing demands for intelligence and efficiency in today's built environment, heating, ventilation, and air conditioning (HVAC) systems—being a critical component of building operations—have received growing attention for their optimized management and efficient maintenance. Building Information Modeling (BIM) provides detailed information about buildings and systems, offering strong support during the planning, design, and construction phases. It enables all project stakeholders to communicate efficiently and make precise decisions within a visualized and collaborative platform, thereby improving overall project execution efficiency and quality.

In response to these challenges and opportunities, this study proposes a BIM-based digital twin framework designed to support the intelligent operation and maintenance of central air-conditioning systems. The framework leverages real-time data integration and secondary development of BIM models to enable dynamic, data-driven management. Unlike previous research that treats BIM, digital twins, or HVAC systems in isolation, this work presents an integrated platform that combines these technologies into a unified and practical solution.

The paper is structured as follows: Section 2 introduces the literature review; Section 3 introduces the digital twin framework required for the platform; Section 4 details the modeling and secondary development of laboratory equipment; Section 5 implements real-time data communication for the platform; Section 6 presents the design, development, and functional modules of the digital twin platform; Section 7 tests the platform's performance and presents results; Section 8 discusses the research findings; The final section concludes the study.

2. Literature review

In recent years, Building Information Modeling (BIM) and digital twin (DT) technologies have gained increasing attention in the fields of architecture, energy, and facility management. Their integration not only enhances building energy efficiency and intelligent operations but also shows broad application prospects in energy system optimization, indoor environmental control, and industrial system modeling. Studies have demonstrated that digital twins, when combined with AI and IoT, enable real-time monitoring, fault diagnosis, and performance optimization of HVAC systems. Meanwhile, the collaboration with BIM supports dynamic control and efficient management across the building lifecycle. Existing literature has proposed BIM-based HVAC optimization models, indoor thermal comfort prediction systems, and systematic reviews, highlighting the importance of standardization and data interoperability in building effective BIM–DT platforms. Building on this foundation, the present study focuses on the integration strategies and practical value of BIM and digital twins in addressing real-time changes in HVAC system usage.

In the field of energy system optimization, digital twins are used to coordinate and control multi-vector smart energy systems. O'Dwyer et al. [6] integrated energy management tools with digital twins to achieve dynamic coordination of smart energy systems, significantly improving sustainability. Butean et al. [7] proposed a digital twin framework for optimizing high-temperature heat pumps, reducing energy waste in industrial renewable energy systems. Bani-Hani et al. [8] combined sustainable refrigerants with digital twins to develop an energy-efficient and eco-friendly solution for machine tool cooling systems. In the building environmental control domain, digital twin technology enhances the intelligence of building energy management. Cui et al. [9] developed a digital twin model for PERA's air conditioning system, optimizing performance through real-time monitoring and simulation. Manfren et al. [10] demonstrated the application of digital twins in building energy monitoring using smart thermostatic radiator valves and gas absorption heat pumps. Lu et al. [11] introduced a clustering compression-based calibration method for HVAC digital twins, reducing computational complexity in modeling. ElArwady et al. [12] integrated digital twins with machine learning to establish a high-precision indoor thermal comfort prediction model, supporting dynamic environmental regulation. In industrial system modeling and optimization, digital twins are widely adopted for complex system performance enhancement. Yang et al. [13] built a compressor characteristic surrogate model using digital twins and proposed optimization algorithms to improve efficiency. Cui et al. [14] developed a data-driven digital twin model for supercritical coal-fired boilers, enabling early detection of metal temperature anomalies. Tao et al. [15] proposed a five-dimensional digital twin model and systematically summarized its technical framework and ten major applications, providing methodological guidance for industrial intelligence.

Research on digital twin technology in building energy management, industrial system optimization, and indoor environmental control continues to deepen, with applications gradually expanding to HVAC systems. In HVAC systems, digital twin technology is widely used for fault detection, model updates, and performance optimization. Xie et al. [16] developed a digital twin-based fault diagnosis method to improve equipment maintenance efficiency. Chen et al. [17] utilized a broad learning system (BLS) algorithm to achieve dynamic updates of HVAC digital twin models. Ma et al. [18] combined sparse autoencoders (SSAE) with transfer learning to build a chiller fault diagnosis digital twin model, enhancing anomaly detection robustness. Digital twin technology, integrated with IoT and AI, drives dynamic monitoring and intelligent regulation of indoor environments. Zhang et al. [19] proposed a digital twin-driven ventilation system for automated humidity optimization in underground heritage sites. Qian et al. [20] designed an intelligent digital twin platform for real-time indoor air quality monitoring and control. Zhang et al. [21] innovatively proposed a digital twin-driven fresh air curtain wall generation method to dilute air pollutants, dynamically protecting historical building facades. Li et al. [22] optimized the performance of cascade refrigeration systems using digital twin models. Machado et al. [23] developed a digital twin model for solar absorption chillers, advancing renewable energy applications in cooling systems. Yoon et al. [24] integrated digital twin technology with ontology through an AI agent-based framework, enabling buildings to achieve real-time operational management.

In building lifecycle management, the integration of BIM and smart building technologies provides more accurate support for real-time monitoring and optimized control of equipment. Alshehri et al. [25] proposed enhancing the sustainability and resilience of the construction industry by combining BIM with emerging technologies such as artificial intelligence and the Internet of Things (IoT).

Alves et al. [26] demonstrated that integrating BIM with AI enables building managers to monitor and regulate all building systems and equipment in real time. Guo et al. [27] introduced a framework that integrates BIM, remote sensing sensors, and IoT for quality monitoring in residential projects. Hu et al. [28] combined BIM, LiDAR, and IoT technologies to achieve real-time monitoring and full-cycle visualization of indoor environments. Hasnain et al. [29] highlighted that the widespread adoption of BIM not only optimizes project execution but also offers significant advantages in cost control, risk management, and resource allocation. When further integrated with technologies like IoT, VR, and AI, BIM demonstrates even greater scalability and forward-looking potential in facility operation and maintenance management.

In the research on the integration of BIM and digital twins, Hosamo et al. [30] proposed an HVAC digital twin model based on a BIM framework, incorporating artificial neural networks and a multi-objective genetic algorithm to optimize the balance between energy consumption and thermal comfort. Jiang et al. [31] developed an optimization framework for hospital HVAC design by integrating BIM and digital twin technologies to address real-time changes in functional usage. To enhance building occupant comfort, Hosamo et al. [32] also developed a digital twin method based on Bayesian networks and predictive maintenance strategies, enabling real-time data-driven analysis of indoor environmental conditions and optimization of HVAC operations. Nguyen and Adhikari [33] conducted a systematic review of the integration patterns of BIM and digital twins in the building construction phase, emphasizing that standardization, data interoperability, and cross-phase integration strategies are key to establishing efficient BIM–DT platforms. Deng et al. [34] systematically evaluated the development trends of BIM and digital twins in intelligent building modeling, data-driven decision-making, and operational automation, and pointed out that a smooth transition from BIM to digital twins relies on enhanced real-time data acquisition, semantic information richness, and platform integration mechanisms.

In summary, digital twin technology enables the prediction, evaluation, and control of air-conditioning systems by facilitating seamless integration and efficient interaction between physical equipment and its digital counterpart. Moreover, the integration of BIM with smart building technologies provides precise support for real-time monitoring and optimized control of equipment, thereby

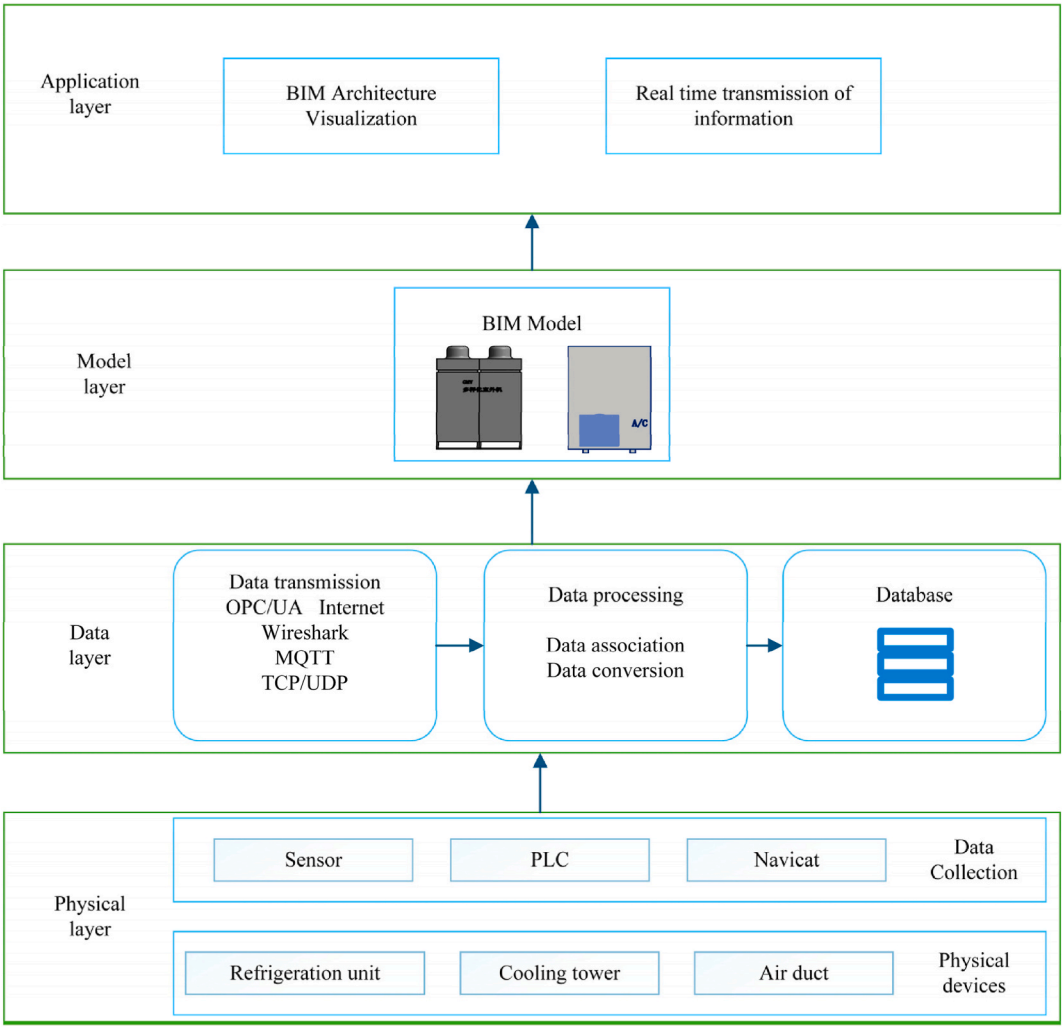


Fig. 1. A system framework for the development of a BIM-based central air conditioning digital twin visualization monitoring and platform.

driving the construction industry toward a more intelligent, efficient, and sustainable future.

The integration of digital twin technology in these studies mainly focuses on the following areas: (1) Visualization during the construction phase: some research applies BIM to early-stage design and structural simulation throughout the building's lifecycle; (2) Equipment fault detection and model updating: such as developing fault detection processes for HVAC systems assisted by digital twins, or implementing self-updating models using broad learning algorithms; (3) Indoor environmental regulation and energy-saving control: for instance, designing platforms for indoor air quality control and proposing compression-based calibration algorithms for HVAC digital twin models; (4) Building energy management: such as conducting multi-dimensional energy efficiency simulations of air-conditioning system performance based on digital twin platforms.

Based on the analysis of the above studies, three major issues remain insufficiently addressed in current research: First, most existing studies focus on individual components within HVAC systems, lacking a system-level modeling and integration framework for the entire central air-conditioning system. Second, the integration of BIM and digital twin technologies is primarily applied to static visualization during the construction phase, failing to resolve the issue of digital silos. Third, there is still a lack of a unified platform architecture that simultaneously ensures the integrity of the BIM model, real-time equipment data, and system scalability. Therefore, the motivation of this study is to extend the application of BIM technology from the design and construction stages to the operation and maintenance phase of central air-conditioning systems, thereby enhancing their intelligent management. The aim of this research is to develop a BIM-based digital twin platform for central air-conditioning, enabling real-time mapping of equipment operation status, data synchronization, and interactive control to support intelligent operation and maintenance. To achieve these goals, the study focuses on the following research questions: (1) How can BIM models be efficiently integrated with digital twin systems? (2) How can a system architecture be designed to ensure scalability, interactivity, and real-time performance?

To address the above issues, this study proposes a modeling and operation platform for central air-conditioning systems that integrates BIM and digital twin technologies. By conducting secondary development of the BIM model for a laboratory air-conditioning system, the study establishes a mapping logic between equipment, model, and data, and designs a data communication mechanism based on the OPC/UA protocol along with a visualized platform architecture. This platform fills the research gap in digital twin modeling for central air-conditioning systems; it builds a data-driven logic structure that supports the fusion of multi-source heterogeneous data and real-time communication, thereby enhancing the platform's scalability and general applicability. Systematic functional and performance testing further validates the platform's stability and reliability, laying a solid foundation for its future engineering deployment in building energy management.

3. Proposed framework

With the proliferation of digital twin technology, this study proposes a BIM-based digital twin framework for central air conditioning systems, based on the established five-dimensional digital twin model framework [15]. The framework synergistically integrates expertise in sensor technology, communication technology, and digital signal processing to develop a highly efficient and accurate BIM-based digital twin platform. As shown in Fig. 1, the system architecture comprises four layers: Physical Layer, Data Layer, Model Layer, and Application Layer, which interact closely through collaborative support, providing a solid technical foundation for smart operation, maintenance, and precise control of central air conditioning systems.

3.1. Physical Layer

As the foundational layer of the framework, the Physical Layer comprises physical equipment such as refrigeration units, air handling units, cooling towers, ducts, and air outlets. It generates data from physical entities and their related subsystems, utilizing sensors and communication protocols for data collection and processing. Physical devices operate based on predefined start-stop requirements, while sensors capture multi-source heterogeneous data (e.g., device attributes, operational status, process metrics, and fault signals). After preliminary processing, the data is transmitted to the next layer. Additionally, the physical systems must respond to commands or feedback issued by the digital twin platform. In this framework, a gateway acts as middleware that connects the Physical Layer and Data Layer through standardized protocols. The gateway converts machine-specific protocols into standard communication protocols to ensure seamless data flow throughout the architecture.

3.2. Data Layer

The Data Layer comprises three core components: data transmission, data processing, and database management. For data transmission, it employs a wide variety of communication protocols and flexible transmission methods to ensure stable and efficient data transfer between diverse devices and systems. In the data processing phase, specialized equipment and sensors are used to perform in-depth analysis of raw collected data, promptly detect anomalies, and process faulty data, thereby ensuring data accuracy and reliability. The database, serving as the storage core, encompasses a broad spectrum of data types, including digital twin data generated during equipment operation, long-term accumulated experiential knowledge, and physics-based model data. When selecting database management software, relational or non-relational systems—such as SQL Server, MySQL, MongoDB, and Redis—can be chosen based on actual needs to meet the storage and management requirements of data with varying scales and structures.

3.3. Model layer

The core of the Model Layer is the BIM model developed using Autodesk Revit software. This model accurately simulates detailed information of various equipment in the mechanical room through parametric equipment modeling and family library construction. The Revit family library provides robust flexibility, enabling rapid adjustments to device models and specifications (e.g., cooling towers, chillers) based on actual requirements. For equipment with no existing families available, users can develop custom families tailored to specific needs, ensuring that the BIM model closely aligns with the operational conditions of physical equipment. This process thereby establishes an accurate digital model foundation for subsequent simulation analyses and operational maintenance management.

3.4. Application Layer

The Application Layer integrates core functions such as 3D visual monitoring of equipment status, real-time start/stop control, and predictive maintenance decision optimization. By utilizing technologies including HTML, CSS, JavaScript, WebGL, Unity3D, and MQTT, this layer achieves real-time, comprehensive monitoring and precise control of equipment through the synergy of twin data and BIM models. Additionally, it provides robust data support and technical assurance for remaining useful life (RUL) prediction and data-driven predictive maintenance strategies, thereby facilitating a transition from reactive to proactive maintenance. This significantly improves operational efficiency and reliability, reduces maintenance costs, and enhances the system's intelligent capabilities and sustainable development potential.

Compared with existing research, this framework demonstrates three innovations: First, it proposes a universal data communication structure based on the OPC/UA protocol, which addresses the issue of static data silos in traditional BIM models during the operation and maintenance phase, enabling real-time bidirectional interaction and mapping between the physical and model layers. Second, the framework can simultaneously integrate BIM data and sensor data, establishing a data fusion channel specifically designed for central air-conditioning systems. Third, the framework includes modules such as real-time equipment status visualization and start-stop control, offering a high degree of integration and practical applicability.

4. Building information modeling and secondary development

BIM technology is essentially a digital building technology that offers future buildings the opportunity to revolutionize contemporary design and operations and maintenance, with the greatest value being in threading through the entire life cycle of a building [35]. To achieve this goal, data interfaces are established through secondary development techniques in order to export electromechanical models to the BIM software, thus completing the in-depth design of the components. Referring to the study by Wang [36] which explored detailed component design methods in prefabricated buildings based on BIM technology, it provided theoretical guidance for the parametric representation and component modeling of equipment elements within the BIM model used in this

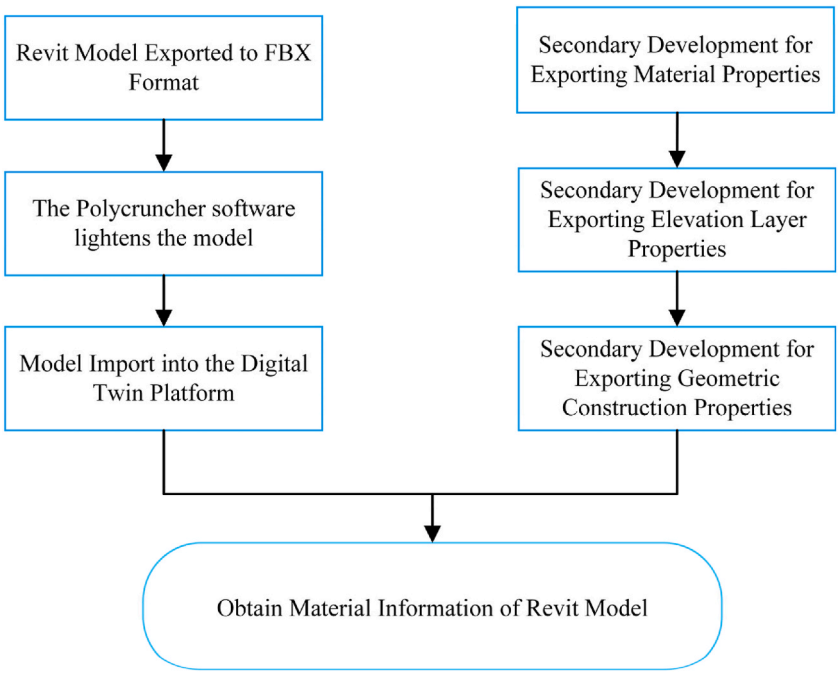


Fig. 2. BIM model secondary development process.

research. The specific secondary development process was as follows: first, the Revit model was exported to FBX format and then the model was lightened using Polycruncher software. Next, the model information was exported using Python scripts. At the same time, the material information of the model is extracted by Visual Studio development program. After the combination of the two, the complete model information can be obtained. The specific process is shown in Fig. 2.

Some of the material information obtained through secondary development is shown in Table 1 (the Air Handling Unit (AHU), chiller, and cooling tower). The Scroller Controller script file was integrated into Unity to enable model functionalities, and material information can appear by clicking on the model. The main flow of the design is as follows: family library design → BIM modelling → development of the corresponding program using VS → model lightweighting → material quantity statistics. The functional display determined in this paper consists of three levels: (1) Design of elevation and category. (2) Material family library with ID design. (3) Export of material information for design finalization and report generation. Fig. 3 shows the BIM module in the digital twin platform. The secondary development of the BIM design supports the expansion of multiple types of equipment, while the OPC/UA communication mechanism enables integration with devices from different manufacturers. Additionally, the data-driven logic can be flexibly adjusted according to the attributes of various equipment. Therefore, this framework is not only suitable for the visualization and operation control of individual components but can also be extended to the full-chain modeling and management of more complex central air-conditioning systems.

5. Real-time mapping of data for central air-conditioning twin model systems

5.1. Data requirements for virtual-real interaction in central air-conditioning

The central air conditioning mechanical room, as a complex and critical system, generates data with distinct multi-source, real-time, and heterogeneous characteristics. The underlying building data are categorized into three types: First, there are BIM-based building entities and equipment data, which provide stable static data for the system; Second, there are the real-time operating data of the internal building equipment, which change with the operation of the equipment and constitute dynamic data; Third, there are other equipment parameters and maintenance data, which include both static inherent parameters of the equipment and dynamic parameters due to maintenance, ensuring that the system comprehensively reflects the operational status of the building. As shown in Fig. 4. Among them, building materials are categorical attribute data (non-numeric), used to identify the material of components; building height is continuous numerical data, which can be used for elevation modeling analysis; and building type is discrete enumerated data, used to define the functional category of equipment or space.

5.2. OPC/UA and communication function based on central air-conditioning system

In digital twin systems, efficient and accurate data exchange and deep integration are key to achieving intelligent and refined control. With the increasing complexity of device types, data structures, and communication scenarios, there is a pressing need for a unified communication protocol and technical framework that can handle multi-source heterogeneous data and support real-time bidirectional communication [37]. Among the many industrial protocols available, OPC/UA is widely regarded as one of the most representative solutions due to its high scalability, security, and cross-platform compatibility.

To illustrate the advantages of OPC/UA technology, consider its application in central air-conditioning systems. The OPC/UA server captures key environmental parameters in real-time and with high-precision, including temperature variations, humidity fluctuations, air quality changes, and detailed equipment status and performance metrics during operation. Through the OPC/UA protocol, these data are securely, rapidly, and accurately transmitted across the real-digital boundary to the digital twin platform, these data thereby serve as a robust driver for system simulation and predictive analysis. UA clients are deployed across applications, devices, and systems to receive and process server data in real time. This enables remote monitoring and intelligent, precise regulation of the central air-conditioning system, ensuring optimal operational performance at all times.

5.3. Structure for real-time mapping of central air-conditioning data

In the structure of the digital twin platform for central air-conditioning, the communication interface and data acquisition module drive the efficient operation of the whole system and are key components in the operation. Fig. 5 shows the system architecture of the digital twin system, which comprises four modules closely connected and synergistic modules: the data acquisition module of the

Table 1
Partial mechanical and electrical equipment material information.

Name	Elevation	Category	Family	Family type	Material
Enthalpy exchanger	2	Mechanical equipment	Enthalpy exchanger	Enthalpy exchanger	Aluminum plate
Chiller Unit	1	Mechanical Equipment	Chiller Unit	Chiller Unit	Stainless Steel
Fresh Air Unit	1	Mechanical equipment	Fresh Air Unit	Air Handling Unit	Stainless Steel
Air Handling Unit	2	Mechanical equipment	Conditioning Indoor Unit	AC Equipment	Copper-Coated Galvanized Steel
Water Pump	1	Mechanical equipment	Water Pump	Water Pump	Cast Iron
Cooling Tower	1	Mechanical equipment	Cooling Tower	Cooling Tower	Galvanized Steel

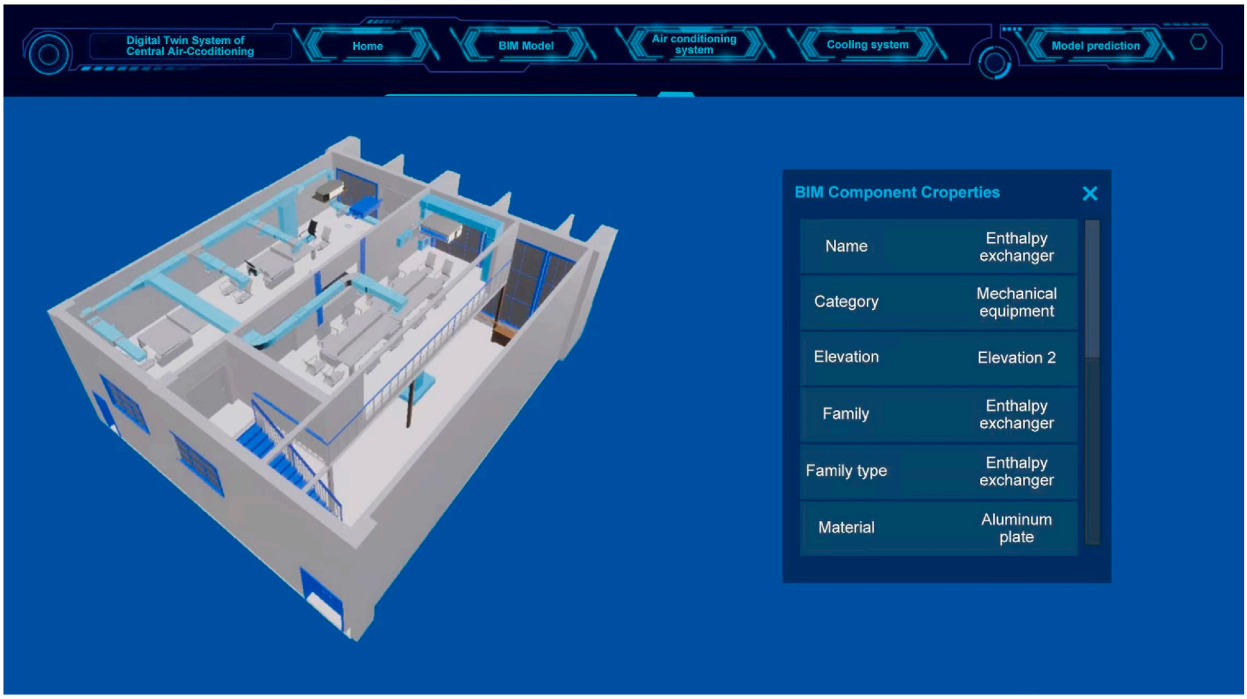


Fig. 3. BIM module of the Central Air-Conditioning Digital Twin Platform.

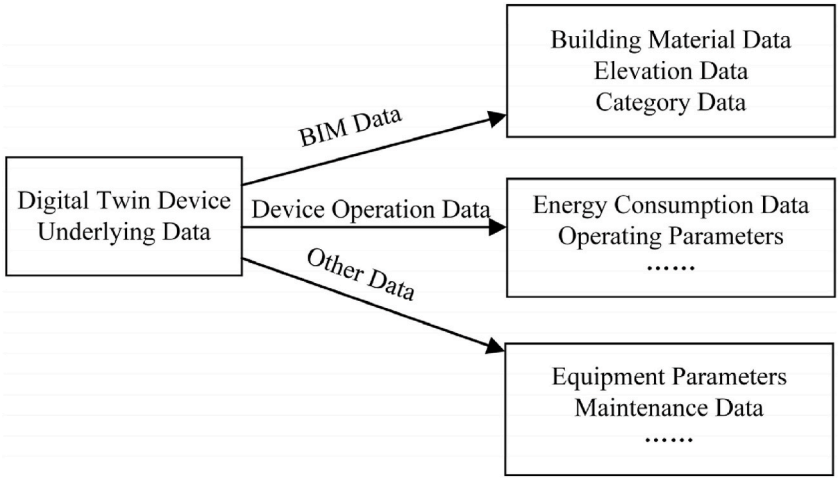


Fig. 4. Device low-level data.

underlying equipment, the control system module, the digital space module, and the data service platform.

This paper also proposes a data-driven logical structure for the central air-conditioning digital twin system. It consists of real-time data input, system initialization and startup, digital space, system state mapping, and control signal feedback channel. These components work in close synergy to create a highly efficient, intelligent, and accurate digital twin system for central air-conditioning. As shown in Fig. 6.

In the digital space, the twin system operates multiple algorithms in parallel. the system operation process is divided into the following steps demonstrated the superiority of the DT-BIM platform in user interaction, response speed, and security:

(1) Data acquisition: In actual operation, sensors are required to acquire data in real time and upload the data to the database for storage.
(2) Real-time data mapping: After the data reaches the large database, the display data is matched with the digital data, and the paired data is uploaded to the twin platform.

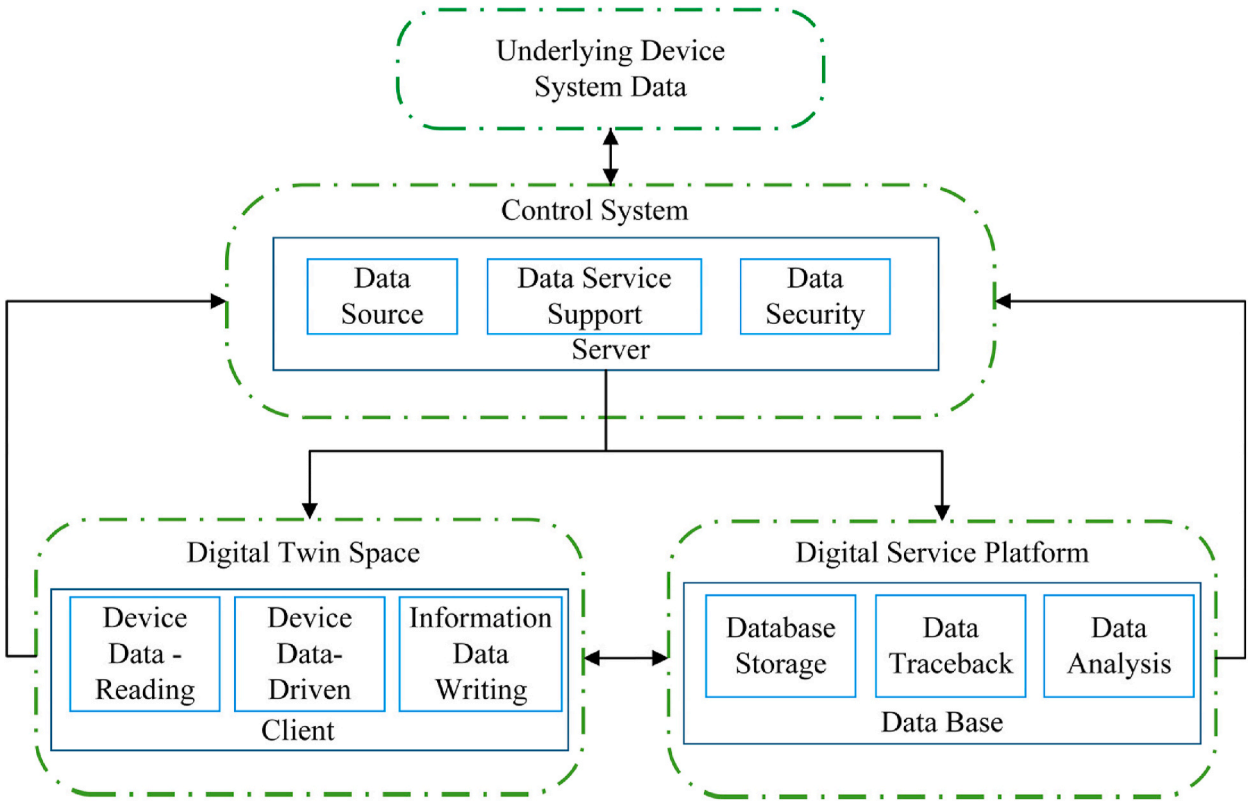


Fig. 5. Digital twin system communication structure.

(3) Data processing: Analyzing and preserving various types of data when mapping the digital space to the real space.
(4) Data service request: First, the PLC is set up to read the required data at regular intervals. Through the code, the data is read and uploaded to the database, where it is stored. Based on the platform's request, a data query is performed in the database, and then the data is sent to the digital twin platform.

For the actual needs of central air-conditioning equipment, the output data primarily consists of real-time operational data for various equipment within the building. The data requests can be divided into the following two types:

(1) Data sent from PLC in real time: In PLC, data can be read and sent to the database, where it is converted into the WSTRING format and uploaded to the database.
(2) Unity extracts the data stored in the database: In Unity, C# code is used to extract the required data into Unity, first declaring a 'public string' variable and then connecting to the database.

6. The design of a digital twin platform for central air conditioning systems

The model developed in this study was conducted with the support of the Artificial Environment Laboratory at Changsha University of Science & Technology. This laboratory features state-of-the-art facilities and a comprehensive control system, providing an ideal foundation for designing the digital twin platform for central air-conditioning systems. Next, we detail the platform's design process. Fig. 7 illustrates the laboratory's internal environment.

In the selection of the development platform for the project, based on the development language and BIM model data, this paper chooses Microsoft Visual Studio 2019 and Revit 2018. The platform is created in Unity and using API interfaces to obtain data and upload the data read by the PLC to the database, this enables operations such as data reading and processing. The overall process is shown in Fig. 8.

By designing the UI of the digital twin platform and converting the BIM model's geometric data through secondary development, this paper successfully developed a digital twin platform for central air conditioning systems. The functional buttons bar is located at the top of the interface, including the Home Page, BIM Display, Air Conditioning System, Cold Source System, and other functions.

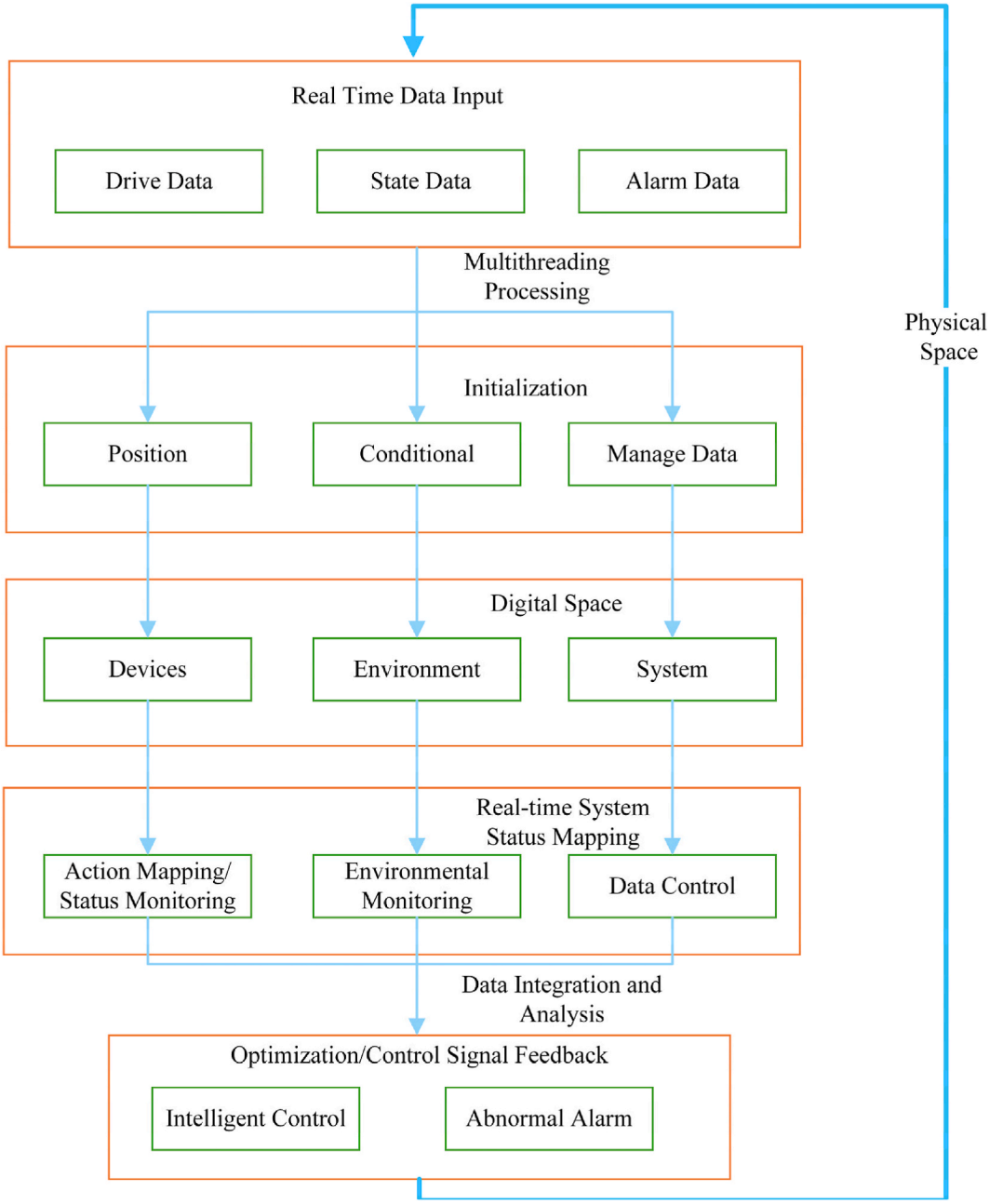


Fig. 6. Digital twin central air conditioning data-driven logic.

6.1. BIM module

Through this interface, users can observe the various devices included in the model within the digital twin platform. Additionally, users can interactively rotate the model to view it from different angles, providing a comprehensive perspective. Furthermore, users can directly click on any device in the model to immediately view its detailed parameters. This feature greatly enhances the accessibility and intuitiveness of the information.

6.2. Air conditioning Air Handling System/Water system section

This article will use the Air Handling System as an example to delve into its functions. This section provides a detailed depiction of the equipment layout of the laboratory's ventilation system. In the figure, each piece of equipment is accompanied by a button. Users



Fig. 7. Laboratory interior environment.

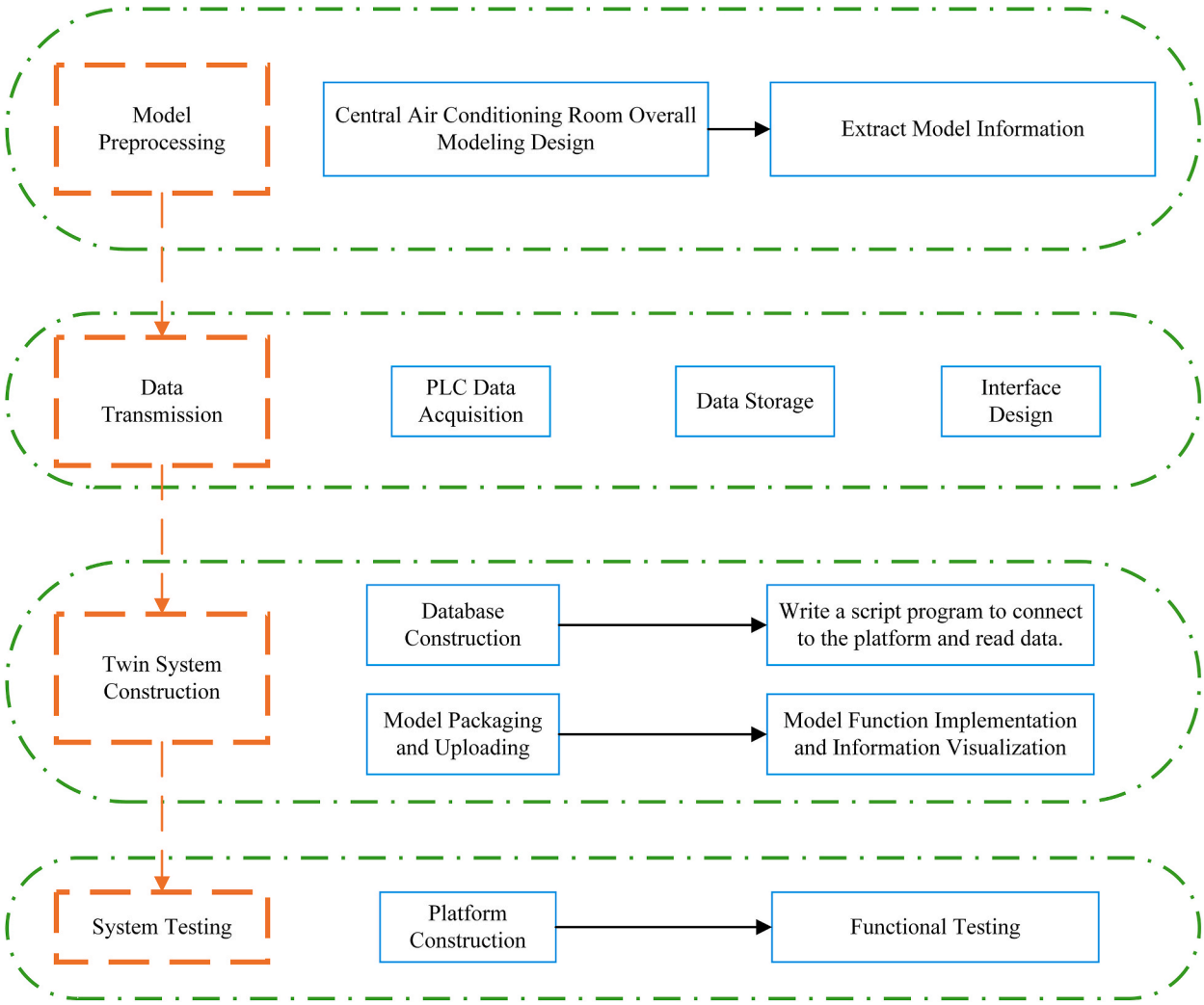


Fig. 8. Platform construction process.

can easily access the equipment's parameter interface with a single click. These parameter interfaces can reflect the latest data in the database in real time, enabling users to intuitively grasp the dynamic changes in equipment status.

6.3. Statistics chart section

A data visualization function has been designed to display key parameters in an intuitive chart format. Specifically, this section will aggregate and analyze the average indoor temperature data for each month. Additionally, a detailed humidity statistics chart is provided, which shows changes in humidity at different times, helping users to better understand the environmental conditions. Such a design not only enhances the readability of the data but also provides users with a convenient means of environmental monitoring. At the same time, this section also collects data on the electricity consumption of the equipment and creates relevant charts.

6.4. Equipment start/stop section

This section is designed within the air system module. Through this button, it allows operators to control the start and stop of equipment via a simple interface. In the air system module, this function is particularly important because the air system typically involves large equipment such as fans and compressors. The start and stop of these devices require precise control to ensure the safety and efficiency of the system.

7. Digital twin platform system performance testing

Performance testing is an essential part of the software development process, playing a crucial role in ensuring the quality and stability of the system. It not only helps identify potential hidden defects but also effectively enhances the overall reliability of the software [38]. The testing was conducted in a Windows 11 operating system environment. The hardware platform used included an Intel Core i9-14900HX processor, 32 GB of DDR5 memory, and an NVIDIA RTX 4060 graphics card. The platform was developed based on Unity 2021 and a MySQL database, and the system was deployed within a local area network (LAN). All tests were carried out under a stable network connection, with Unity Profiler used as the primary testing tool. To ensure the stable operation of the central air conditioning digital twin platform, this section will conduct performance tests on the system's modules, including both functional and non-functional tests.

7.1. Functional testing of the system

Functional testing includes the user login function, including test cases related to accounts and login. Conduct multiple rounds of testing for different input scenarios, as shown in Table 2. Besides, Functional testing also includes the platform's functionality, including interface loading test cases and test cases for BIM model navigation, clicking on models, etc., as shown in Table 3.

7.2. Non-functional testing of the system

Non-functional testing includes aspects such as load performance, security, and reliability. Testing is conducted on performance load and security to ensure that the system achieves stable response times and strong security.

Performance Load Testing: This involves testing interface response times and interface load. The Unity built-in Profiler tool is used to test hardware like CPU and memory. Security Testing: This mainly includes the secure storage of user passwords in encrypted form and the verification of interface permissions. The specific details are shown in Table 4.

8. Discussion

Building upon the five-dimension digital twin model proposed by Tao et al. [15], this study further develops a digital twin framework specifically for central air-conditioning systems. This model systematically defines five core components: physical entities, virtual entities, connections, data, and services, thereby providing a theoretical foundation for the structural modeling of digital twin systems. In the modeling process of this platform, the layered design concept of the five-dimension model was integrated to construct a scalable digital twin framework tailored for central air-conditioning systems. This framework not only supports the digital transformation of such systems but also lays a solid architectural foundation for future platform expansions, offering valuable insights for the intelligent upgrading of other industrial systems as well. Systematic testing, including functional and non-functional assessments, demonstrates the superiority of the DT-BIM platform in user interaction, response speed, and security, laying a practical foundation for its engineering promotion and optimization of building energy management.

Compared to existing research, this study achieves deep synergy between BIM and digital twins. By breaking through the limitation of traditional studies where BIM serves merely as a visualization tool [28], we realize dynamic integration of BIM models with the digital twin platform was realized through secondary development, enabling real-time mapping of equipment status to provide high-fidelity data support for precise control. The proposed OPC/UA protocol-based general-purpose data-driven logical architecture for central air conditioning systems supports efficient processing and synchronization of multi-source heterogeneous data, resolving compatibility issues in existing methods [6] caused by over-reliance on specific software frameworks, thereby enhancing system scalability and standardization. For the first time, platform testing incorporated both functional (e.g., interface interaction, control

Table 2
User login test cases.

Number	Test Case Name	Preconditions	Test Case Procedure	Expected Result	Test
T001	Login Correctly	Enter correct account and password	Click Login	Show login successful	Pass
T002	Account Error	Enter the wrong account	Click to Log In	Display “Incorrect Username or Password"	Pass
T003	Password Incorrect	Enter the wrong password	Click to Log In	Display “Incorrect Username or Password"	Pass

Table 3
User information initialization test case.

Number	Test Case Name	Preconditions	Test Case Procedure	Expected Result	Test
T004	User Information Initialization Interface Display	User Successfully Logged In	Redirect to Basic Information Settings Interface After Login	Correctly Display Saved Basic Information	Pass
T005	Click the BIM Model Button	User Successfully Redirected	Click	Display the BIM Model Interface	Pass
T006	Incorrect Data Format	Click on the Blank Space	Click	No Information Popped Up	Pass

Table 4
Performance load and security test cases.

Number	Test Case Name	Preconditions	Test Case Procedure	Expected Result	Test
T007	CPU Frame Time	None	Platform Execution	Frame time does not exceed 10 ms	Pass
T008	Memory Usage Testing	None	On the running platform, view the average usage situation.	The average memory usage should not exceed 2 GB.	Pass
T009	Rendering Performance	None	Enable the Rendering module and record the CPU’s rendering performance data.	The number of rendering batches should not exceed 2K.	Pass
T010	Script Execution Time	None	Enable Scripts and record the script execution time.	Measure the execution time of the script each frame to ensure it does not exceed 10 ms.	Pass
T011	Password Storage	Create database	Check the database data.	Password Encrypted Storage	Pass

logic) and non-functional (e.g., response time, security) metrics, filling the gap in existing studies of focusing on single performance validation in similar studies.

Despite these interim progresses, limitations require further investigation. The current model exhibits limited adaptability to dynamic scenarios, as its predictive ability under extreme operating conditions or sudden failures has not been validated. Future work should integrate reinforcement learning to improve system adaptability. Additionally, cross-system generalizability requires validation. While the framework currently focuses on central air conditioning systems, its applicability to other complex energy systems (e.g., district cooling/heating networks) requires empirical validation.

9. Conclusion

This paper proposes a BIM-based digital twin platform for central air conditioning systems and demonstrated its feasibility through functional testing and non-functional testing. The conclusions are as follows:

(1) Through theoretical and analytical studies, BIM model parameters and equipment specifications for central air conditioning systems were obtained. A BIM model and family library based on room equipment parameters were constructed, with through secondary development to extract equipment information. This provides the digital twin platform with precise visualization models and accurate device data.
(2) A BIM-based digital twin platform for central air conditioning systems was established, achieving real-time data mapping based on OPC/UA. This created an efficient bidirectional channel for real-time data transmission in central air conditioning systems. The proposed platform enables real-time monitoring of equipment status, provides technical support for fault detection and predictive maintenance.
(3) Functional and non-functional testing ensured the secure and stable operation of the platform, providing a technical framework for smart O&M of central air conditioning systems.

This study makes the following key contributions: First, the proposed platform addresses a significant research gap in the digital twin modeling of central air-conditioning systems. Second, it establishes a robust data-driven logic structure that supports the integration of multi-source heterogeneous data and facilitates real-time communication, thereby enhancing the platform’s scalability and versatility. Finally, the platform has been rigorously validated through comprehensive functional and performance testing, providing a solid foundation for its future implementation in building energy management.

Despite achieving preliminary results, this study still faces several limitations that warrant further exploration. One key issue is the lack of adaptability to dynamic scenarios—the current model's ability to predict extreme conditions or unexpected faults has not yet been validated, and future work should incorporate reinforcement learning to enhance system adaptability. In addition, the cross-system generalizability of the framework remains to be tested, as it is currently focused on central air-conditioning systems. Its applicability to other complex energy systems, such as district cooling or heating networks, requires further evaluation.

Future research can further enhance the development of this platform in several directions: 1. Explore adaptive control strategies based on reinforcement learning to improve the platform's responsiveness to dynamic loads and unexpected events. 2. Expand the validation of the platform's applicability in cross-system scenarios (such as district cooling/heating networks) to assess its performance in more complex environments. 3. Investigate deeper integration of BIM, artificial intelligence (AI), and digital twin technologies to enable intelligent optimization throughout the building design and construction processes.

CRediT authorship contribution statement

Nie Ma: Writing – original draft, Visualization, Validation, Software, Methodology. **Wei Li:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Changwei Jiang:** Writing – review & editing, Supervision, Resources, Project administration. **Xiaoqin Sun:** Writing – review & editing, Visualization, Conceptualization. **Jili Zhang:** Writing – review & editing, Resources, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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